

Exercise Sheet #4

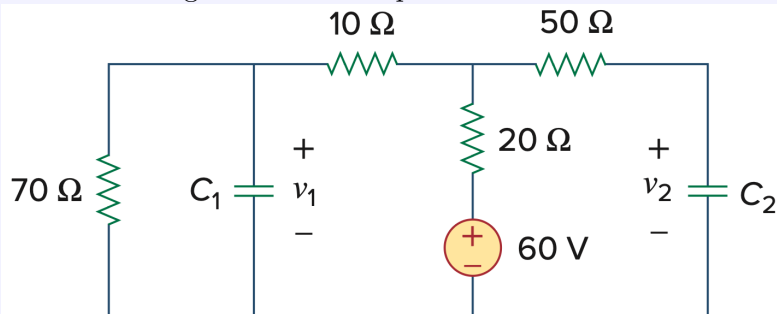
Fields 1

Exercise 1: Discrete Charge Distribution

Two point charges of $+3.0 \mu\text{C}$ each are located at the points $(x, y) = (0, 2.0 \text{ m})$ and $(x, y) = (0, -2.0 \text{ m})$. Two additional point charges, each with charge q , are located at $(x, y) = (4.0 \text{ m}, 2.0 \text{ m})$ and $(x, y) = (4.0 \text{ m}, -2.0 \text{ m})$. The electric field at the point $(x, y) = (0, 0)$ due to the four charges is $\vec{E} = (4.0 \times 10^3 \hat{x}) \text{ N/C}$. Determine the value of q .

Exercise 2: Capacitors in DC Steady State

Find the voltages across the capacitors in the circuit below under DC conditions.

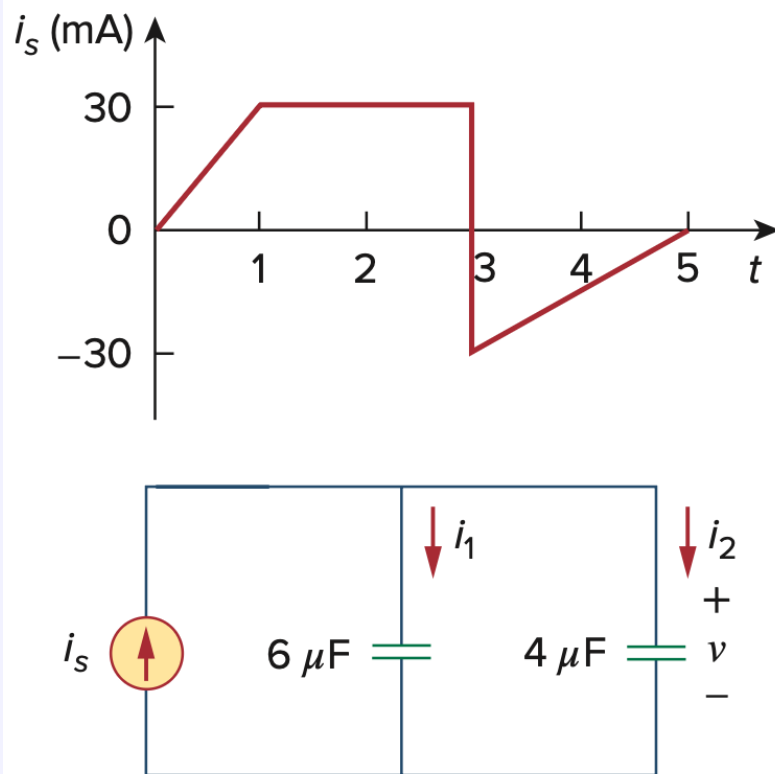


Exercise 3: Plate Capacitor

We want to design a capacitor with capacitance $C = 1 \mu\text{F}$. Compute the length required for rectangular plates of 2 cm width if the dielectric is polyester ($\epsilon_r = 3.4$) of $15 \mu\text{m}$ thickness.

Exercise 4: Time-Dependent Capacitor Voltage and Current

If $v(0) = 0$, find $v(t)$, $i_1(t)$, and $i_2(t)$ in the circuit below.



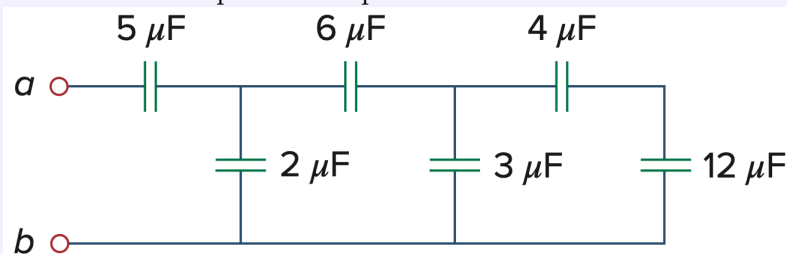
Exercise 5: Energy of Capacitors

Two capacitors ($25 \mu\text{F}$ and $75 \mu\text{F}$) are connected to a 100 V source. Find the energy stored in each capacitor if they are connected in:

- a) parallel
- b) series

Exercise 6: Series and Parallel Capacitors

Determine the equivalent capacitance at terminals a-b of the circuit below.



Exercise 7: Magnetic Force

In a straight, current-carrying section of a conductor there is a current element

$$I\vec{l}$$

with

$$I = 2.7 \text{ A}, \quad \vec{l} = (3.0 \hat{x} + 4.0 \hat{y}) \text{ cm}.$$

It is surrounded by a uniform magnetic field

$$\vec{B} = 1.3 \hat{x} \text{ T.}$$

Calculate the force acting on the section of the conductor.

Exercise 8: Lorentz Force

A velocity selector operates with a magnetic field of $B = 0.28 \text{ T}$ perpendicular to an electric field of $E = 0.46 \text{ MV/m}$. How fast must a particle move in order to pass through the selector without deflection?

Exercise 9: Faraday's Law: Moving Conductor

A rectangular conducting loop has width $l = 0.2 \text{ m}$ (the side that crosses the field boundary) and height $h = 0.1 \text{ m}$. The loop lies initially entirely inside a region of uniform magnetic field B directed *into* the page, with magnitude $B = 0.8 \text{ T}$. The loop is pulled to the right with constant speed $v = 0.5 \frac{\text{m}}{\text{s}}$, so that it leaves the magnetic region.

- Calculate the magnitude of the induced voltage in the loop while it is partially leaving the magnetic region.
- State the direction of the induced current (clockwise or counterclockwise) as seen by the observer, and justify using Lenz's law.

Exercise 10: Faraday's Law: Time-varying Field

A circular conducting loop of radius $r = 5 \text{ cm}$ lies in a region where the magnetic field perpendicular to the loop is increasing in time according to

$$B(t) = 0.012t \text{ T} \quad \text{with } t \text{ in seconds.}$$

The loop resistance is $R = 2 \Omega$.

- Find the magnitude of the induced voltage at $t = 4 \text{ s}$.
- Find the induced current magnitude at that time and specify its sense (clockwise or counterclockwise) if B points *out* of the page.